SOC 527

SOCIAL NETWORKS

Stochastic Actor-Oriented Models

Inferential Network Models Guest Lecture Series

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THE UNIVERSITY OF ARIZONA
COLLEGE OF SOCIAL & BEHAVIORAL SCIENCES
School of Sociology

Lecture Materials

Lecture Handout:

Course Notes for SOC 527 (Social Networks)

This is a course handout for the lecture on stochastic actor-oriented models (SAOMs) for SOC 527: Social Networks at the University of Arizona. The course handout provides a brief technical introduction for stochastic actor-oriented models and a (rather incomplete) reference list that details the methodological development of the SAOM. Importantly, this course handout should be read alongside the lecture slide deck and the course R code script. In this way, (1) the slide deck provides a general understanding and example empirical applications of the SAOM, (2) the R code file explains how to estimate an SAOM given empirical data using the RSiena R package, and (3) the course handout provides a more technical introduction for the SAOM that goes beyond the lecture slide deck.

Last Updated: October 19, 2026 Kevin A. Carson (kacarson@arizona.edu)

What is the stochastic actor-oriented model?

The **stochastic actor-oriented model (SAOM)** is an inferential network model that allows us to understand what network- and actor-based processes lead to network changes across waves of data collection. In particular, the **SAOM** was developed with the early work of Snijders (1996, 150) who wanted to create a "statistical model that is itself a direct expression of ... sociological theory" to study the evolution of networks. Within the **SAOM** framework, it is assumed that actors (also known as agents; e.g., adolescents in high schools) are "individually optimizing his or her own utility," where the 'utility' is based upon a set of network and exogenous constraints and the evolution of the network between waves "is the result of the actions of the individual actors in the network" (Snijders, 1996, 152). Moreover, the term **actor-oriented** comes from the idea that actors within the network control their outgoing ties, that is, who they send a tie to, and that "changes in ties are made ... on the basis of their and others' attributes, their position in the network, and their perceptions about the rest of the network" (Snijders, Van De Bunt and Steglich,

Replication R Script:

```
###-----------------------------------------------------------------------------###
#
#       Introduction to Stochastic Actor-Oriented Models
#       University of Arizona School of Sociology SOC 527: Social Networks
#       Code Written by Kevin A. Carson (https://kevincarson.github.io/)
#       Last Updated: 10-08-2026
#
###-----------------------------------------------------------------------------###
rm(list = ls()) #clearing the working memory
library(RSiena) #reading the RSiena package (for the estimation of SAOMs)
library(sna) #reading the sna package (for the computation of clustering)
library(network) #reading the network package
library(netmediate) #reading in the netmediate package (for the computation of MEMs)
library(networkDynamic) #reading the networkDynamic package to store the dynamic microstep movie
library(ndtv)  #reading the ndtv package to make the microstep video object
library(ggplot2) #reading in the ggplot2 to make some nice plots
sessionInfo() #for complete replication
set.seed(3718)

# R version 4.5.2 (2025-10-31)
# Platform: aarch64-apple-darwin20
# Running under: macOS Tahoe 26.6.2
#
# Matrix products: default
# BLAS:   /System/Library/Frameworks/Accelerate.framework/Versions/A/Frameworks/vecLib.framework/Versions/A/libBLAS.dylib
# LAPACK: /Library/Frameworks/R.framework/Versions/4.5-arm64/Resources/lib/libRlapack.dylib;  LAPACK version 3.12.1
#
# locale:
# [1] en_US.UTF-8/en_US.UTF-8/en_US.UTF-8/C/en_US.UTF-8/en_US.UTF-8
#
# time zone: America/New_York
# tzcode source: internal
#
# attached base packages:
# [1] stats     graphics  grDevices utils     datasets  methods   base
#
# other attached packages:
# [1] ggplot2_4.0.1        ndtv_0.13.4          animation_2.8        networkDynamic_0.11.5 netmediate_1.1.1
# [6] sna_2.8              network_1.19.0       statnet.common_4.12.0 RSiena_1.5.0
#
# loaded via a namespace (and not attached):
# [1] Rdpack_2.6.4           lpSolveAPI_5.5.2.0-17.14 rlang_1.1.6            magrittr_2.0.4
# [5] compiler_4.5.2         vctrs_0.6.5              stringr_1.6.0          pkgconfig_2.0.3
# [9] fastmap_1.2.0          labeling_0.4.3           rmarkdown_2.30         purrr_1.2.0
# [13] xfun_0.58              cachem_1.1.0             ergm.multi_0.3.0       trust_0.1-8
```

Both have been uploaded to the course **d2l** site!

Lecture Outline

- Continuous-time markov models for the studying the evolution of social networks
- Stochastic actor-oriented models
- Empirical application



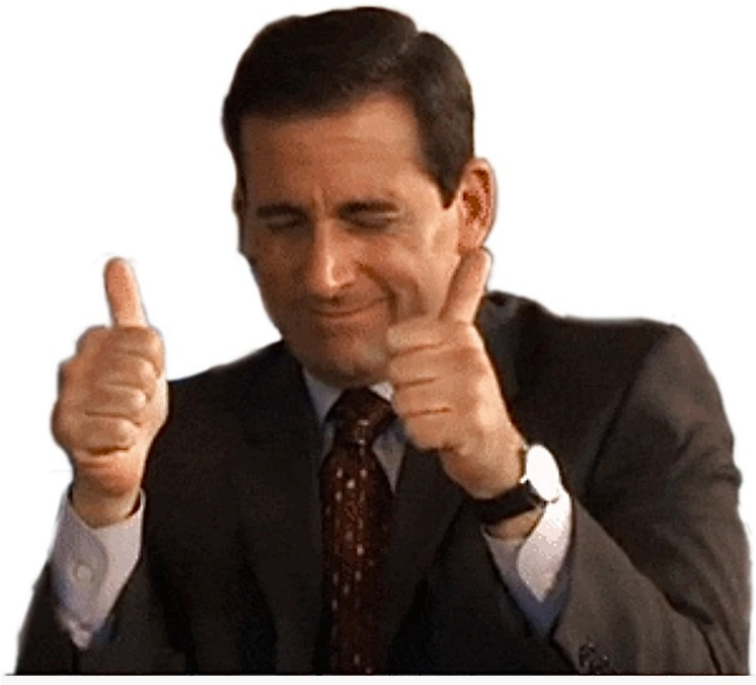
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Lecture Goals

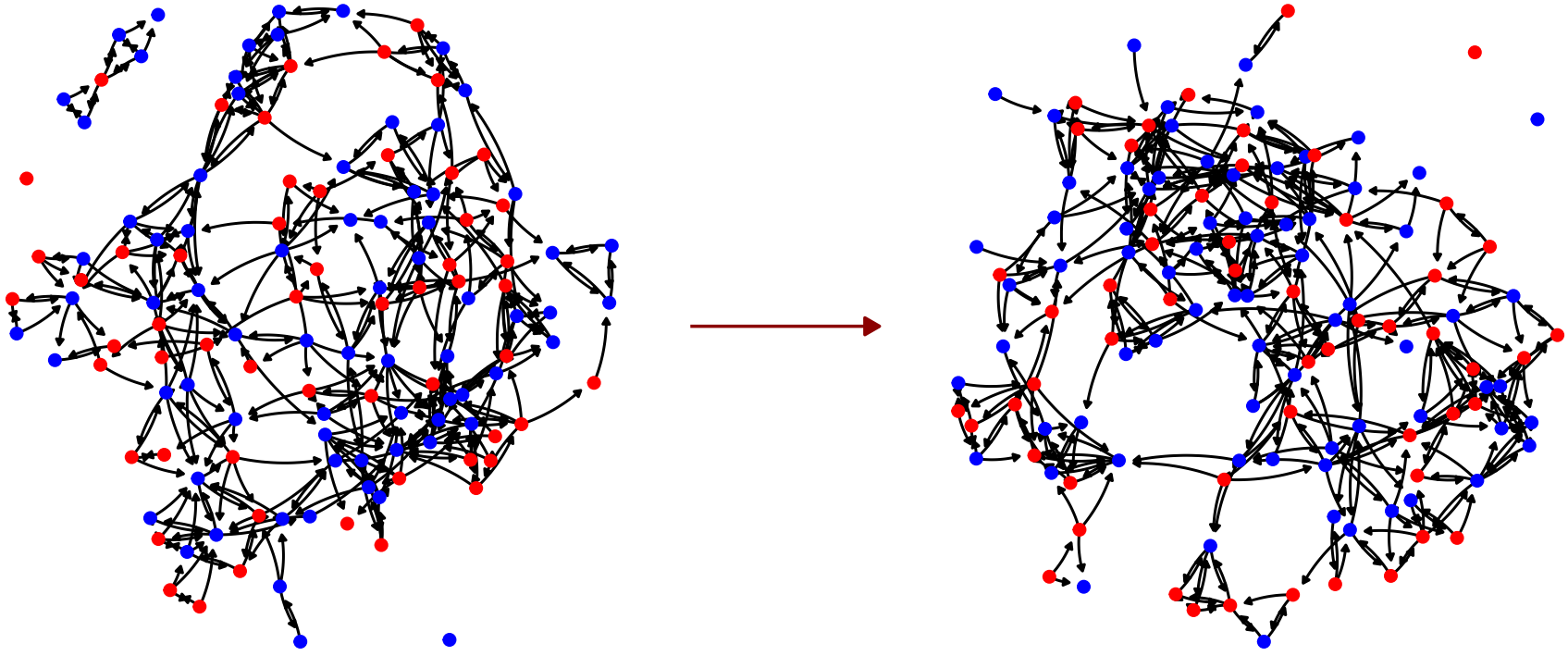
1. Have a general understanding of the stochastic actor-oriented model.

2. Be able to (a) interpret the results and (b) diagnose the goodness of fit of a fitted SAOM.

3. Be able to tell if you can use the SAOM with your observational dataset.



Glasgow Wave 1 **Glasgow Wave 2**



The **stochastic actor-oriented model (SAOM)** is an inferential continuous-time markov model that allows us to understand how a network changes across waves of data collection [Snijders, 1996, Snijders et al., 2010].

Why use the stochastic actor-oriented model?

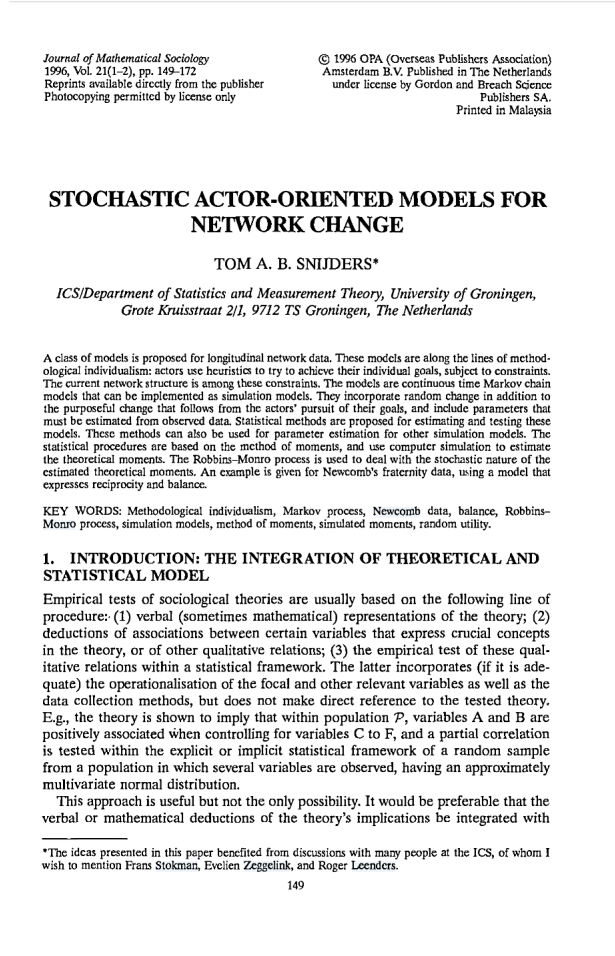
The stochastic actor-oriented model allows us to:

- → **test hypotheses about network change**
- → **simulate networks under different research designs**
- → **predict network structures and ties**



Source: https://freshspectrum.com/evaluation-cartoons/

The SAOM was popularized by the early work of Snijders [1996, 150] who wanted to create a "statistical model that is itself a direct expression of ... sociological theory" to study the evolution of networks.



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STOCHASTIC ACTOR-ORIENTED MODELS FOR NETWORK CHANGE

TOM A. B. SNIJDERS*

ICS/Department of Statistics and Measurement Theory, University of Groningen, Grote Kruisstraat 2/1, 9712 TS Groningen, The Netherlands

A class of models is proposed for longitudinal network data. These models are along the lines of methodological individualism: actors use heuristics to try to achieve their individual goals, subject to constraints. The current network structure is among these constraints. The models are continuous time Markov chain models that can be implemented as simulation models. They incorporate random change in addition to the purposeful change that follows from the actors' pursuit of their goals, and include parameters that must be estimated from observed data. Statistical methods are proposed for estimating and testing these models. These methods can also be used for parameter estimation for other simulation models. The statistical procedures are based on the method of moments, and use computer simulation to estimate the theoretical moments. The Robbins–Monro process is used to deal with the stochastic nature of the estimated theoretical moments. An example is given for Newcomb's fraternity data, using a model that expresses reciprocity and balance.

KEY WORDS: Methodological individualism, Markov process, Newcomb data, balance, Robbins–Monro process, simulation models, method of moments, simulated moments, random utility.

1. INTRODUCTION: THE INTEGRATION OF THEORETICAL AND STATISTICAL MODEL

Empirical tests of sociological theories are usually based on the following line of procedure: (1) verbal (sometimes mathematical) representations of the theory; (2) deductions of associations between certain variables that express crucial concepts in the theory, or of other qualitative relations; (3) the empirical test of these qualitative relations within a statistical framework. The latter incorporates (if it is adequate) the operationalisation of the focal and other relevant variables as well as the data collection methods, but does not make direct reference to the tested theory. E.g., the theory is shown to imply that within population $\mathcal{P}$, variables A and B are positively associated when controlling for variables C to F, and a partial correlation is tested within the explicit or implicit statistical framework of a random sample from a population in which several variables are observed, having an approximately multivariate normal distribution.

This approach is useful but not the only possibility. It would be preferable that the verbal or mathematical deductions of the theory's implications be integrated with

*The ideas presented in this paper benefited from discussions with many people at the ICS, of whom I wish to mention Frans Stokman, Evelien Zeggelink, and Roger Leenders.

Source: Snijders [1996]

The SAOM was popularized by the early work of Snijders [1996, 150] who wanted to create a "statistical model that is itself a direct expression of ... sociological theory" to study the evolution of networks.

→ **Methodological individualism**

- ☞ In this theory, agents are purposive actors who choose social actions that maximize their 'utility' based upon a set of constraints.
- ☞ more on this later!

→ **Continuous-time markov models**

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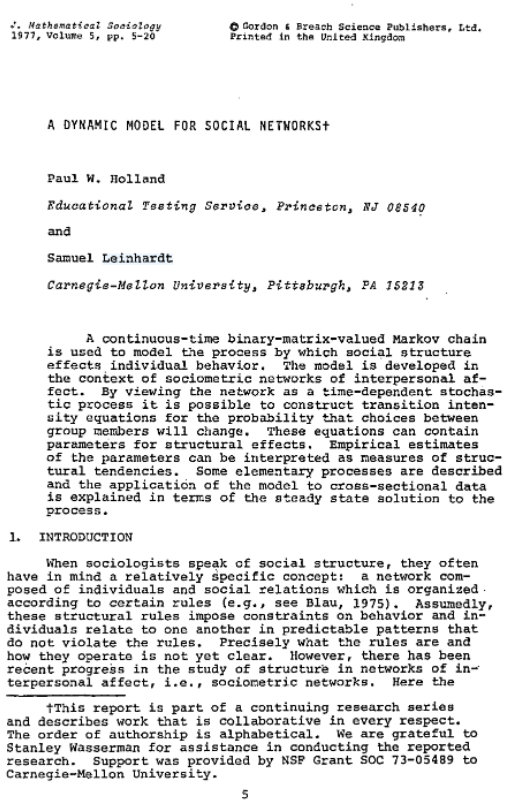
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Source: Snijders [1996]

Some Preliminaries: Continuous-time markov models for the evolution of social networks

- **Continuous-time markov models** for social network dynamics were popularized by the early work of Holland and Leinhardt [1977], Wasserman [1980], and Leenders [1995].



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A DYNAMIC MODEL FOR SOCIAL NETWORKS†

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and

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A continuous-time binary-matrix-valued Markov chain is used to model the process by which social structure effects individual behavior. The model is developed in the context of sociometric networks of interpersonal affect. By viewing the network as a time-dependent stochastic process it is possible to construct transition intensity equations for the probability that choices between group members will change. These equations can contain parameters for structural effects. Empirical estimates of the parameters can be interpreted as measures of structural tendencies. Some elementary processes are described and the application of the model to cross-sectional data is explained in terms of the steady state solution to the process.

1. INTRODUCTION

When sociologists speak of social structure, they often have in mind a relatively specific concept: a network composed of individuals and social relations which is organized according to certain rules (e.g., see Blau, 1975). Assumedly, these structural rules impose constraints on behavior and individuals relate to one another in predictable patterns that do not violate the rules. Precisely what the rules are and how they operate is not yet clear. However, there has been recent progress in the study of structure in networks of interpersonal affect, i.e., sociometric networks. Here the

†This report is part of a continuing research series and describes work that is collaborative in every respect. The order of authorship is alphabetical. We are grateful to Stanley Wasserman for assistance in conducting the reported research. Support was provided by NSF Grant SOC 73-05489 to Carnegie-Mellon University.

Some Preliminaries: Continuous-time markov models for the evolution of social networks

Following Holland and Leinhardt [1977], let $X(t)$ be a continuous-time stochastic process representing a $g \times g$ binary sociomatrix where social ties among g actors evolve over time.

$$X(t) = \begin{bmatrix} 0 & X_{12}(t) & X_{13}(t) & \cdots & X_{1g}(t) \\ X_{21}(t) & 0 & X_{23}(t) & \cdots & X_{2g}(t) \\ X_{31}(t) & X_{32}(t) & 0 & \cdots & X_{3g}(t) \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ X_{g1}(t) & X_{g2}(t) & X_{g3}(t) & \cdots & 0 \end{bmatrix} \tag{1}$$

where

$$X_{ij}(t) = \begin{cases} 1 & \text{if } i \to j \text{ at time } t \\ 0 & \text{otherwise} \end{cases} \tag{2}$$

Some Preliminaries: Continuous-time markov models for the evolution of social networks

- Following Holland and Leinhardt [1977], let $X(t)$ be a continuous-time stochastic process representing a $g \times g$ binary sociomatrix where social ties evolve over time with g actors.
  - $$X_{ij}(t) = \begin{cases} 1 & \text{if } i \to j \text{ at time } t \\ 0 & \text{otherwise} \end{cases} \tag{3}$$
  - $t \in \mathbb{R}$ is a continuous time parameter with $t \geq 0$ [Leenders, 1995].

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  - $t \in \mathbb{R}$ is a continuous time parameter with $t \geq 0$ [Leenders, 1995].
  - $X(t) = x$ is the realized sociomatrix at time t [Holland and Leinhardt, 1977].

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  - $$X_{ij}(t) = \begin{cases} 1 & \text{if } i \to j \text{ at time } t \\ 0 & \text{otherwise} \end{cases} \tag{5}$$
  - $t \in \mathbb{R}$ is a continuous time parameter with $t \geq 0$ [Leenders, 1995].
  - $X(t) = x$ is the realized sociomatrix at time t [Holland and Leinhardt, 1977].
  - S is the state space that contains all possible $g \times g$ binary sociomatrices with $x, y, z, k, \cdots$ being possible sociomatrices.

Some Preliminaries: How does $X(t)$ evolve over time?

Assumption 1. $X(t)$ is a Markov process, such that, the distribution of future network state $X(t+h)$ in a small time interval h is dependent upon the current network state x [Wasserman, 1980]:

$$\mathrm{P}_{x \to y}(t, h) = \mathrm{P}\{X(t+h) = y | X(t) = x\} \tag{6}$$

- h represents a small interval of time $(t, t+h)$.
- As Holland and Leinhardt [1977, 8] asserted: "we wish to reflect the idea that the structure of the network at time t can influence the probability of subsequent changes in its structure."

Some Preliminaries: How does $X(t)$ evolve over time?

Assumption 2. Given the current network state, x, at time t, subsequent changes in the ties are statistically independent of each other after a small amount of time [Holland and Leinhardt, 1977]:

$$P_{x \to y}(t, h) = \prod_{i,j} P\{X_{ij}(t+h) = y_{ij} | X(t) = x\} + o(h) \quad \text{as} \quad h \to 0 \tag{7}$$

- $o(h)$ is a function that goes to zero faster than h.
- This assumption implies that the probability of more than one tie changing simultaneously within the interval is of order $o(h)$, which is negligible in comparison to one tie changing as the time interval $h \to 0$ [Holland and Leinhardt, 1977].

Some Preliminaries: How does $X(t)$ evolve over time?

The **change intensities**, λ, for the process for a small time interval $(t, t+h)$ is:

$$\underbrace{\mathrm{P}\{X_{ij}(t+h) = 1 - x_{ij} | X(t) = x\}}_{\text{probability that the } i \to j \text{ tie changes}} = h\lambda_{ij}(x,t) + o(h)$$

Some Preliminaries: How does $X(t)$ evolve over time?

The change intensities are typically characterized as a **function of the current network state**, x:

$$h\lambda_{ij}(x,t) = h \cdot \underbrace{f(x)}_{\text{statistics that capture the current network state}} \tag{8}$$

- Reciprocity model: Is the $i \to j$ tie more likely to be created (dropped) if the $j \to i$ tie exists? [Wasserman, 1980]
- Popularity model: Is the $i \to j$ tie more likely to be created (dropped) if j has received many ties? [Wasserman, 1980]

Methodological individualism and the actor-oriented framework

In Methodological individualism and the **SAOM** it is assumed that actors are "individually optimizing his or her own utility" [Snijders, 1996, 152].

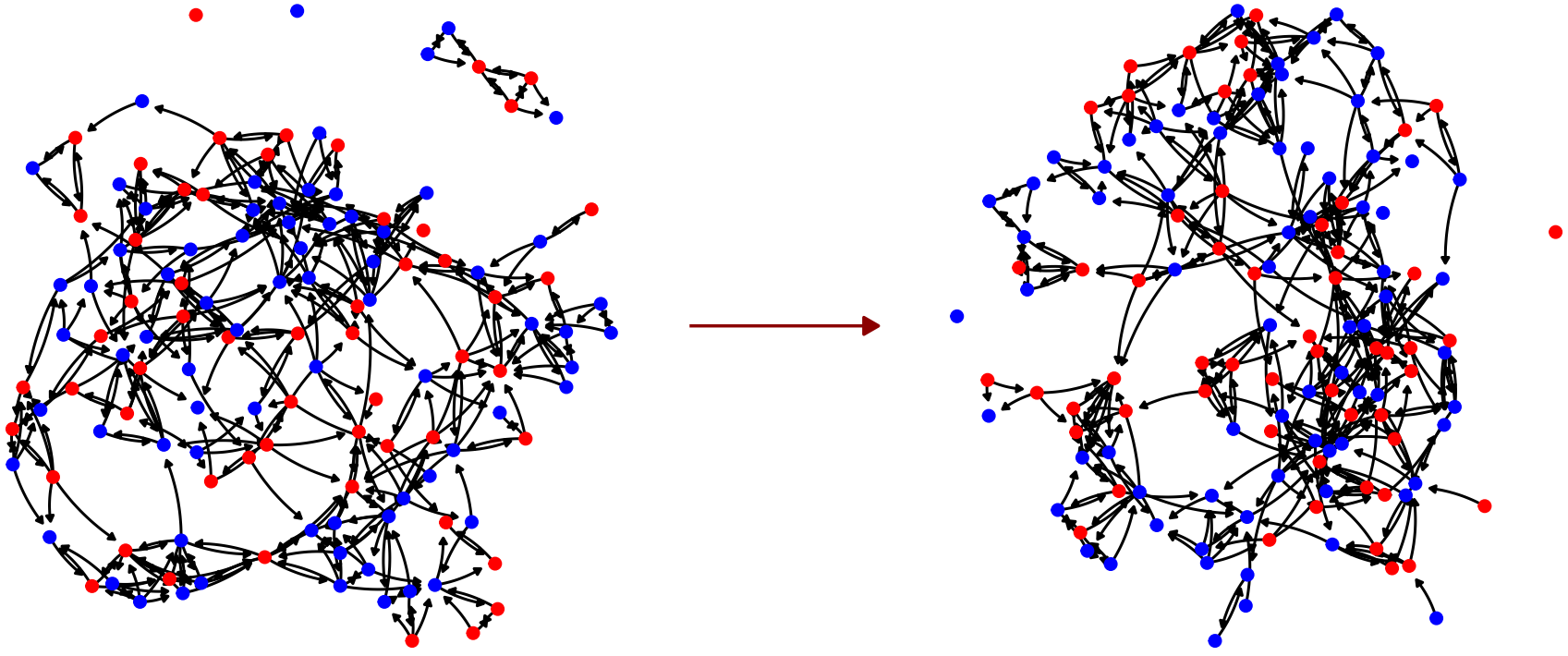
- for the SOAM, the 'utility' is based upon a set of network and exogenous constraints.
- actors within the network control their outgoing ties and "changes in [their] ties are made ... on the basis of their and others' attributes, their position in the network, and their perceptions about the rest of the network" [Snijders et al., 2010, 46].

Stochastic actor-oriented model assumptions

In the SAOM it is assumed that [Snijders, 2001]:

- actors have knowledge of the current network state x and the distribution of actor attributes.
- actors control their outgoing ties and change their ties based upon the maximization of their 'utility' function.
- changes in the network state follow a Markov process with changes in the subsequent network being conditionally independent in small time intervals h.
- only one actor at a time has the opportunity to change their outgoing ties.

What is the actor-oriented framework?



Based on the actor-oriented assumptions, the SAOM assumes that the changes in the network structure between data collection waves are the result of an underlying latent process whereby changes are the "result of the actions of the individual actors in the network" [Snijders, 1996, 152].

How does the SAOM model the evolution of the network?

The SAOM is a two-step **change determination continous time markov model** where changes to the network state occur through a sequence of 'microsteps':

- In the first step, actors are randomly chosen to get the opportunity to change a tie.
- In the second step, the chosen actor evaluates their current outgoing ties and make a choice about changing or not changing their ties.

Step 1. the opportunity to change a tie

In the first step, actors are randomly chosen to get the opportunity to change a tie. The waiting time until an actor i is chosen to make a change to their outgoing ties is exponentially distributed with rate parameter $\lambda_i(x, z)$ [Snijders, 1996, 2001].

- ➞ The rate parameter, $\lambda_i(x, z)$, may be based upon the current network state x or a node-level attribute z.
  - ☞ Leal et al. [2026] test if the rate parameter is a function of adolescent depression (z).
- ➞ Commonly, we assume that the rate parameter λ_i is homogeneous for all actors.

Step 1. the opportunity to change a tie

The probability that an actor i is selected to make a change at the current microstep is ρ_i [Steglich and Snijders, 2022, Duxbury, 2023]:

$$\rho_i = \frac{\lambda_i(x, y)}{\sum_j \lambda_j(x, y)} \tag{9}$$

Assuming homogeneity of the rate parameter for g actors [Snijders, 2001, Duxbury, 2023]:

$$\rho_i = \frac{1}{g} \tag{10}$$

Step 2. evaluating the current network state

In the second step, the chosen actor i evaluates their current outgoing ties and makes a choice about changing or not changing their ties based upon maximizing their **objective function**.

$$f_i(\beta, x(i \rightsquigarrow j)) \tag{11}$$

- $x(i \rightsquigarrow j)$ denotes the toggling of the x_{ij} tie in the x sociomatrix.
  - if $x_{ij} = 1$, then $x_{ij}(i \rightsquigarrow j) = 0$ (the dropping of the tie).
  - if $x_{ij} = 0$, then $x_{ij}(i \rightsquigarrow j) = 1$ (the creation of the tie).
- The objective function f_i "indicates the degree of satisfaction for actor i inherent in the relational situation represented by" $x(i \rightsquigarrow j)$ [Snijders, 2001, 366].

Step 2. evaluating the current network state

The objective function is a linear combination of parameters β and effects $s_{ik}(x(i \rightsquigarrow j))$ [Snijders, 2001, Duxbury, 2023].

$$f_i(\beta, x(i \rightsquigarrow j)) = \sum_k \beta_k s_{ik}(x(i \rightsquigarrow j)) \tag{12}$$

→ β_k represents the 'importance' of the network statistic s_{ik} for the objective function.

Step 2. evaluating the current network state

$s_{ik}(x(i \rightsquigarrow j))$ are effects that represent a network statistic based upon the network $x(i \rightsquigarrow j)$ [Snijders, 2001]. Let $x^0 = x(i \rightsquigarrow j)$:

- Reciprocity effect: $s(x^0) = \sum_j x^0_{ij} x^0_{ji}$
- Indegree popularity effect: $s(x^0) = \sum_j x^0_{ij} \sum_k x^0_{kj}$
  - $\sum_k x^0_{kj}$ is the indegree of the actor j who i is related to.
- Outdegree activity effect: $s(x^0) = \sum_j x^0_{ij} \sum_k x^0_{jk}$
  - $\sum_k x^0_{jk}$ is the outdegree of the actor j who i is related to.
- Transitivity effect: $s(x^0) = \sum_j \sum_h x^0_{ij} x^0_{ih} x^0_{jh}$
- Actor outdegree (ego) effect: $s(x^0) = y_i \sum_j x^0_{ij}$, where y_i is the covariate value of y for i.

Step 2. evaluating the current network state

The probability that the $i \rightarrow j$ tie changes during the microstep is:

$$p_{ij}(\beta, x) = \frac{\exp(f_i(\beta, x(i \rightsquigarrow j)))}{\sum_k \exp(f_i(\beta, x(i \rightsquigarrow k)))} \tag{13}$$

- The denominator is the set of all possible tie changes, including the network state for which the actor i makes no changes to their outgoing ties, $x(i \rightsquigarrow j) = x$.
- The above formulation is equivalent to the McFadden choice model in economics [Snijders, 2001].

Step 2. evaluating the current network state

β_k are in **log-probability ratios**!

→ Per the RSiena manual, β_k is the "contributions to log-probabilities of increasing the dependent variable by 1 unit when the effect is increased by 1 unit" [Ripley et al., 2026, 218].

Log-Probabilty Ratios:

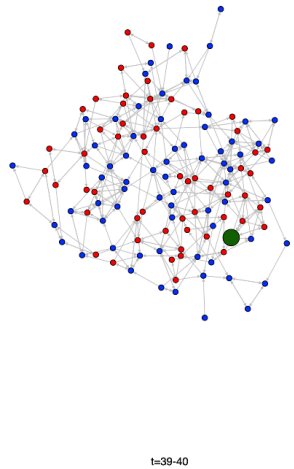
$$\ln\left(\frac{p_{ik}(\beta_k, x)}{p_{ij}(\beta_k, x)}\right) = \beta_k\{s_{ik}(x(i \rightsquigarrow j)) - s_{ik}(x(i \rightsquigarrow z))\} \qquad (14)$$

How does the network state evolves across microsteps?

The probability of a specific $i \rightarrow j$ tie changing during a microstep (i.e., a small interval of time $t + h$) is:

$$\rho_i \cdot p_{ij}$$

It is often helpful to visualize the 'microsteps' of a stochastic actor-oriented model [adams and Schaefer, 2018]!



Click **here** to watch the microstep movie!

Method of Moments Estimation

Snijders [1996] and Snijders [2001] proposed to use Method of Moments estimation to estimate the set of β parameters. Let

- s be a $k \times 1$ vector of statistics from the observed network in wave m (for two waves, $m = 2$).
- $E_\beta(S)$ be a $k \times 1$ vector of statistics from the simulated network in wave m based upon the fitted model.

We find the $k \times 1$ β vector that solves the moment equation:

$$E_\beta(S) = s \tag{15}$$

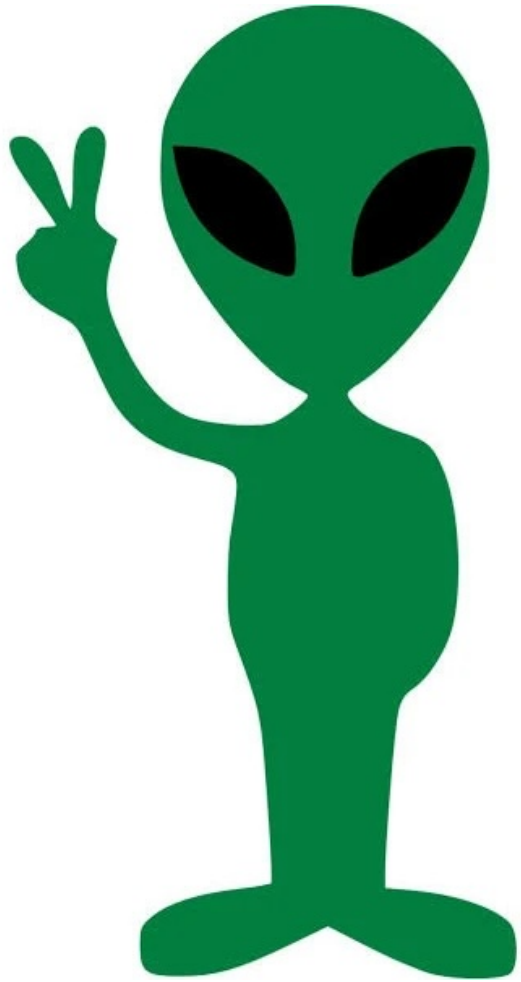
Method of Moments Estimation Routine

The SAOM solves the moment equation via a three-phased computer simulation! [Snijders, 1996]

- The computer simulation use a Robbins–Monro Monte Carlo Markov Chain estimation algorithm.
- For sake of brevity, conditional upon the first network, the SAOM simulates a Monte Carlo Markov Chain of sequential microsteps and updates the parameter estimates based upon the distance between the observed statistics s and the simulated statistics $E_\beta(S)$.
- For sake of brevity, see Snijders [1996], Snijders [2001], and Ripley et al. [2026] for more information on the algorithm.

Method of Moments Estimation Routine

In this regard, the SAOM is commonly described as an agent-based model that allows for statistical inference!



The RSiena R Package

The **RSiena** package is a publicly available and extensively developed package in the R statistical computing environment for the estimation and simulation of stochastic actor-oriented models.

```
#install.packages(RSiena)
library(RSiena)
```

Empirical Application: Glasgow Data

The Glasgow dataset is based upon Teenage Friends and Lifestyle Study, in which adolescents where tracked over their second, third, and fourth year of secondary school in Glasgow. In our examples, we examine the 129 students that present at each wave of data collection and *only examine the second and third year waves*.

Empirical Application: Glasgow Data

Attributes and Networks:

- Friendship Networks: Adolescents where asked if the other students are a best friend, just a friend, or not a friend. We code the network such that:

$$X_{ij}(t) = \begin{cases} 1 & \text{if } i \text{ nominated } j \text{ as a (best) friend at time } t \\ 0 & \text{otherwise} \end{cases} \tag{16}$$

- Node-level attributes:
  - Gender: 1 = boys; 0 = girls
  - Smoking: 1 (non), 2 (occasional) and 3 (regular).

SAOM Results of Glasgow Friendship Dynamics

```
## Estimates, standard errors and convergence t-ratios
##
##                                          Estimate   Standard   Convergence
##                                                       Error      t-ratio
##
## Rate parameters:
##   0        Rate parameter                 11.0409  ( 0.9898   )
##
## Other parameters:
##    1. eval outdegree (density)           -0.7149  ( 0.4684   )   -0.0034
##    2. eval reciprocity                    2.4310  ( 0.1673   )   -0.0063
##    3. eval transitive triplets            0.7909  ( 0.0679   )    0.0496
##    4. eval transitive recipr. triplets   -0.4639  ( 0.1003   )    0.0361
##    5. eval indegree - popularity (sqrt)   0.3088  ( 0.1123   )    0.0180
##    6. eval outdegree - popularity (sqrt) -0.7599  ( 0.2013   )    0.0154
##    7. eval outdegree - activity (sqrt)   -0.7487  ( 0.1292   )    0.0220
##    8. eval smoking alter                  0.0103  ( 0.1006   )   -0.0361
##    9. eval smoking ego                   -0.0343  ( 0.1288   )   -0.0652
##   10. eval smoking similarity             0.5154  ( 0.2003   )    0.0667
##   11. eval gender alter                   0.0389  ( 0.1269   )   -0.0418
##   12. eval gender ego                    -0.0522  ( 0.1620   )    0.0074
##   13. eval same gender                    0.8560  ( 0.1309   )    0.0259
##
## Overall maximum convergence ratio:    0.2309
##
##
## Total of 4041 iteration steps.
```

SAOM Results of Glasgow Friendship Dynamics

```
## Estimates, standard errors and convergence t-ratios
##
##                                        Estimate   Standard   Convergence
##                                                     Error      t-ratio
##
## Rate parameters:
##   0        Rate parameter                11.0409  ( 0.9898   )
##
## Other parameters:
##    1. eval outdegree (density)           -0.7149  ( 0.4684   )  -0.0034
##    2. eval reciprocity                    2.4310  ( 0.1673   )  -0.0063
##    3. eval transitive triplets            0.7909  ( 0.0679   )   0.0496
##    4. eval transitive recipr. triplets   -0.4639  ( 0.1003   )   0.0361
##    5. eval indegree - popularity (sqrt)   0.3088  ( 0.1123   )   0.0180
##    6. eval outdegree - popularity (sqrt) -0.7599  ( 0.2013   )   0.0154
##    7. eval outdegree - activity (sqrt)   -0.7487  ( 0.1292   )   0.0220
##    8. eval smoking alter                  0.0103  ( 0.1006   )  -0.0361
##    9. eval smoking ego                   -0.0343  ( 0.1288   )  -0.0652
##   10. eval smoking similarity             0.5154  ( 0.2003   )   0.0667
##   11. eval gender alter                   0.0389  ( 0.1269   )  -0.0418
##   12. eval gender ego                    -0.0522  ( 0.1620   )   0.0074
##   13. eval same gender                    0.8560  ( 0.1309   )   0.0259
##
## Overall maximum convergence ratio:    0.2309
##
##
## Total of 4041 iteration steps.
```

A fitted SAOM is said to converge when:

- The maximum overall convergence ratio is less than 0.25 (our value: 0.2309) and
- all individual t-ratios are less than 0.10 in absolute value (max value: 0.0667).

SAOM Results of Glasgow Friendship Dynamics

```
## Estimates, standard errors and convergence t-ratios
##
##                                        Estimate   Standard   Convergence
##                                                     Error      t-ratio
##
## Rate parameters:
##   0       Rate parameter                11.0409  ( 0.9898   )
##
## Other parameters:
##    1. eval outdegree (density)          -0.7149  ( 0.4684   )   -0.0034
##    2. eval reciprocity                   2.4310  ( 0.1673   )   -0.0063
##    3. eval transitive triplets           0.7909  ( 0.0679   )    0.0496
##    4. eval transitive recipr. triplets  -0.4639  ( 0.1003   )    0.0361
##    5. eval indegree - popularity (sqrt)  0.3088  ( 0.1123   )    0.0180
##    6. eval outdegree - popularity (sqrt) -0.7599  ( 0.2013   )    0.0154
##    7. eval outdegree - activity (sqrt)   -0.7487  ( 0.1292   )    0.0220
##    8. eval smoking alter                 0.0103  ( 0.1006   )   -0.0361
##    9. eval smoking ego                  -0.0343  ( 0.1288   )   -0.0652
##   10. eval smoking similarity            0.5154  ( 0.2003   )    0.0667
##   11. eval gender alter                  0.0389  ( 0.1269   )   -0.0418
##   12. eval gender ego                   -0.0522  ( 0.1620   )    0.0074
##   13. eval same gender                   0.8560  ( 0.1309   )    0.0259
##
## Overall maximum convergence ratio:    0.2309
##
##
## Total of 4041 iteration steps.
```

The rate parameter is the estimated number of ministeps an actor was chosen to make a change.

→ On average, actors had around 11 opportunities to make a change to their outgoing ties.

SAOM Results of Glasgow Friendship Dynamics

```
## Estimates, standard errors and convergence t-ratios
##
##                                        Estimate   Standard   Convergence
##                                                     Error      t-ratio
##
## Rate parameters:
##   0        Rate parameter                11.0409  ( 0.9898   )
##
## Other parameters:
##    1. eval outdegree (density)           -0.7149  ( 0.4684   )   -0.0034
##    2. eval reciprocity                    2.4310  ( 0.1673   )   -0.0063
##    3. eval transitive triplets            0.7909  ( 0.0679   )    0.0496
##    4. eval transitive recipr. triplets   -0.4639  ( 0.1003   )    0.0361
##    5. eval indegree - popularity (sqrt)   0.3088  ( 0.1123   )    0.0180
##    6. eval outdegree - popularity (sqrt) -0.7599  ( 0.2013   )    0.0154
##    7. eval outdegree - activity (sqrt)   -0.7487  ( 0.1292   )    0.0220
##    8. eval smoking alter                  0.0103  ( 0.1006   )   -0.0361
##    9. eval smoking ego                   -0.0343  ( 0.1288   )   -0.0652
##   10. eval smoking similarity             0.5154  ( 0.2003   )    0.0667
##   11. eval gender alter                   0.0389  ( 0.1269   )   -0.0418
##   12. eval gender ego                    -0.0522  ( 0.1620   )    0.0074
##   13. eval same gender                    0.8560  ( 0.1309   )    0.0259
##
## Overall maximum convergence ratio:    0.2309
##
##
## Total of 4041 iteration steps.
```

In general, the outdegree effect will be negative, as realized social networks are more sparse than expected by random chance.

SAOM Results of Glasgow Friendship Dynamics

We can test the 'importance' of an effect using a Wald-type test. Assume that the null hypothesis for β_k is $H_0 : \beta_k = 0$. The test statistic for the hypothesis is:

$$\frac{\hat{\beta}_k - 0}{SE(\hat{\beta}_k)} \tag{17}$$

The test statistic follows an approximate standard normal distribution.

For outdegree, $\hat{\beta}_k = -0.7149$ and $SE(\hat{\beta}_k) = 0.4684$, the test statistic is -1.526 with associated p-value equal to:

```
2*pnorm(abs(0.7149/0.4684),lower.tail = FALSE)
```

```
## [1] 0.1269452
```

SAOM Results of Glasgow Friendship Dynamics

```
## Estimates, standard errors and convergence t-ratios
##
##                                        Estimate   Standard   Convergence
##                                                     Error      t-ratio
##
## Rate parameters:
##   0        Rate parameter                11.0409  ( 0.9898   )
##
## Other parameters:
##    1. eval outdegree (density)          -0.7149  ( 0.4684   )   -0.0034
##    2. eval reciprocity                   2.4310  ( 0.1673   )   -0.0063
##    3. eval transitive triplets           0.7909  ( 0.0679   )    0.0496
##    4. eval transitive recipr. triplets  -0.4639  ( 0.1003   )    0.0361
##    5. eval indegree - popularity (sqrt)  0.3088  ( 0.1123   )    0.0180
##    6. eval outdegree - popularity (sqrt) -0.7599  ( 0.2013   )    0.0154
##    7. eval outdegree - activity (sqrt)   -0.7487  ( 0.1292   )    0.0220
##    8. eval smoking alter                 0.0103  ( 0.1006   )   -0.0361
##    9. eval smoking ego                  -0.0343  ( 0.1288   )   -0.0652
##   10. eval smoking similarity            0.5154  ( 0.2003   )    0.0667
##   11. eval gender alter                  0.0389  ( 0.1269   )   -0.0418
##   12. eval gender ego                   -0.0522  ( 0.1620   )    0.0074
##   13. eval same gender                   0.8560  ( 0.1309   )    0.0259
##
## Overall maximum convergence ratio:    0.2309
##
##
## Total of 4041 iteration steps.
```

Reciprocity: if $\hat{\beta}_k > 0$,actors are more likely to maintain and create ties to alters who have nominated them.

Transitive triplets: if $\hat{\beta}_k > 0$, actors are more likely to maintain and create ties to alters who are friends of a friend.

Transitive recip. triplets: if $\hat{\beta}_k > 0$, actors are less likely to maintain and create ties to alters who are friends of a friend when the ties are reciprocated.

SAOM Results of Glasgow Friendship Dynamics

```
## Estimates, standard errors and convergence t-ratios
##
##                                        Estimate   Standard   Convergence
##                                                     Error      t-ratio
##
## Rate parameters:
##   0        Rate parameter                11.0409  ( 0.9898   )
##
## Other parameters:
##    1. eval outdegree (density)           -0.7149  ( 0.4684   )   -0.0034
##    2. eval reciprocity                    2.4310  ( 0.1673   )   -0.0063
##    3. eval transitive triplets            0.7909  ( 0.0679   )    0.0496
##    4. eval transitive recipr. triplets   -0.4639  ( 0.1003   )    0.0361
##    5. eval indegree - popularity (sqrt)   0.3088  ( 0.1123   )    0.0180
##    6. eval outdegree - popularity (sqrt) -0.7599  ( 0.2013   )    0.0154
##    7. eval outdegree - activity (sqrt)   -0.7487  ( 0.1292   )    0.0220
##    8. eval smoking alter                  0.0103  ( 0.1006   )   -0.0361
##    9. eval smoking ego                   -0.0343  ( 0.1288   )   -0.0652
##   10. eval smoking similarity             0.5154  ( 0.2003   )    0.0667
##   11. eval gender alter                   0.0389  ( 0.1269   )   -0.0418
##   12. eval gender ego                    -0.0522  ( 0.1620   )    0.0074
##   13. eval same gender                    0.8560  ( 0.1309   )    0.0259
##
## Overall maximum convergence ratio:    0.2309
##
##
## Total of 4041 iteration steps.
```

Indegree popularity: if $\hat{\beta}_k > 0$, actors are more likely to maintain and create ties to alters who have higher indegrees (popularity).

Outdegree activity: if $\hat{\beta}_k > 0$, actors who have higher outdegrees are more likely to maintain and create ties to other alters (sociality).

Outdegree popularity: if $\hat{\beta}_k > 0$, actors are more likely to maintain and create ties to alters who have nominated many other actors.

SAOM Results of Glasgow Friendship Dynamics

```
## Estimates, standard errors and convergence t-ratios
##
##                                        Estimate   Standard   Convergence
##                                                     Error      t-ratio
##
## Rate parameters:
##   0        Rate parameter                11.0409  ( 0.9898   )
##
## Other parameters:
##    1. eval outdegree (density)           -0.7149  ( 0.4684   )   -0.0034
##    2. eval reciprocity                    2.4310  ( 0.1673   )   -0.0063
##    3. eval transitive triplets            0.7909  ( 0.0679   )    0.0496
##    4. eval transitive recipr. triplets   -0.4639  ( 0.1003   )    0.0361
##    5. eval indegree - popularity (sqrt)   0.3088  ( 0.1123   )    0.0180
##    6. eval outdegree - popularity (sqrt) -0.7599  ( 0.2013   )    0.0154
##    7. eval outdegree - activity (sqrt)   -0.7487  ( 0.1292   )    0.0220
##    8. eval smoking alter                  0.0103  ( 0.1006   )   -0.0361
##    9. eval smoking ego                   -0.0343  ( 0.1288   )   -0.0652
##   10. eval smoking similarity             0.5154  ( 0.2003   )    0.0667
##   11. eval gender alter                   0.0389  ( 0.1269   )   -0.0418
##   12. eval gender ego                    -0.0522  ( 0.1620   )    0.0074
##   13. eval same gender                    0.8560  ( 0.1309   )    0.0259
##
## Overall maximum convergence ratio:    0.2309
##
##
## Total of 4041 iteration steps.
```

Gender ego: if $\hat{\beta}_k > 0$, males are more likely to send ties. In other words, gender is associated with the oudegree distribution.

Gender alter: if $\hat{\beta}_k > 0$, males are more likely to recieve ties. Gender is associated with the indegree distribution.

Same gender: if $\hat{\beta}_k > 0$, actors are more likely to maintain and create ties to alters who have the same gender.

SAOM Results of Glasgow Friendship Dynamics

```
## Estimates, standard errors and convergence t-ratios
##
##                                        Estimate   Standard   Convergence
##                                                     Error     t-ratio
##
## Rate parameters:
##   0        Rate parameter                11.0409  ( 0.9898   )
##
## Other parameters:
##    1. eval outdegree (density)           -0.7149  ( 0.4684   )   -0.0034
##    2. eval reciprocity                    2.4310  ( 0.1673   )   -0.0063
##    3. eval transitive triplets            0.7909  ( 0.0679   )    0.0496
##    4. eval transitive recipr. triplets   -0.4639  ( 0.1003   )    0.0361
##    5. eval indegree - popularity (sqrt)   0.3088  ( 0.1123   )    0.0180
##    6. eval outdegree - popularity (sqrt) -0.7599  ( 0.2013   )    0.0154
##    7. eval outdegree - activity (sqrt)   -0.7487  ( 0.1292   )    0.0220
##    8. eval smoking alter                  0.0103  ( 0.1006   )   -0.0361
##    9. eval smoking ego                   -0.0343  ( 0.1288   )   -0.0652
##   10. eval smoking similarity             0.5154  ( 0.2003   )    0.0667
##   11. eval gender alter                   0.0389  ( 0.1269   )   -0.0418
##   12. eval gender ego                    -0.0522  ( 0.1620   )    0.0074
##   13. eval same gender                    0.8560  ( 0.1309   )    0.0259
##
## Overall maximum convergence ratio:    0.2309
##
##
## Total of 4041 iteration steps.
```

Same gender: $\hat{\beta}_k = 0.856$.

Imagine i is evaluating creating two outgoing ties $i \to h$ and $i \to j$.

- Assume that i and j have the same gender while h does not.
- All else being equal, the log difference in the objective functions for creating the $i \to j$ tie, in comparison to the $i \to h$ tie, is $\hat{\beta}_k$!

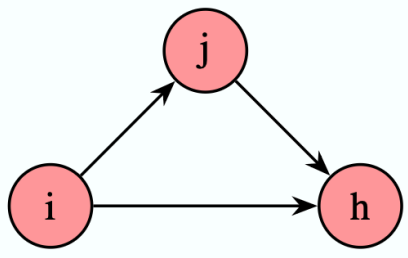
$$\frac{p_{i \to j}}{p_{i \to h}} = \exp[\hat{\beta}_k] = 2.353 \tag{18}$$

How to the specify the effects for an SAOM?

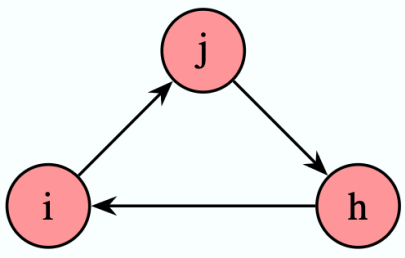
→ Outdegree (density) and reciprocity

How to the specify the effects for an SAOM?

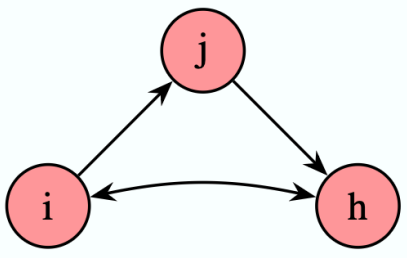
- Outdegree (density) and reciprocity
- Triadic and local structures
  - There are many ways to do this. One way to do this is to include *transitive triplets* and *transitive reciprocated triplets*. [Block, 2015]



(1) Transitive triplets



(2) Three-cycles



(3) Transitive reciprocated triplets

How to the specify the effects for an SAOM?

- Outdegree (density) and reciprocity
- Triadic and local structures
- Heterogeneity in the degree-related distributions
  - *indegree popularity*, *outdegree activity*, and one of the two: *indegree activity*, *outdegree popularity*.
  - *indegree popularity*: the effect of the indegree (number of received ties) for actor j on the probability of forming the $i \rightarrow j$ tie.
  - *outdegree activity*: the effect of the outdegree (number of sent ties) for actor i on the probability of forming the $i \rightarrow j$ tie.

How to the specify the effects for an SAOM?

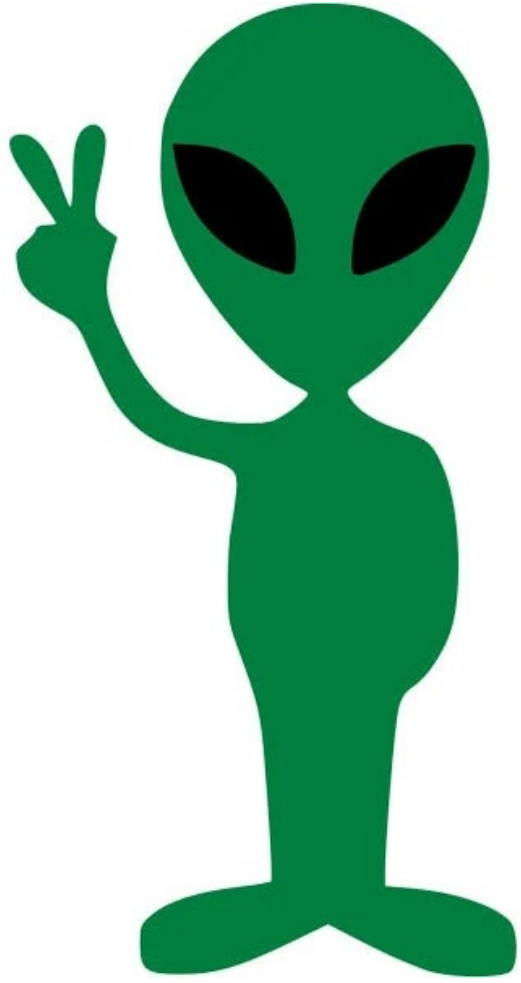
- Outdegree (density) and reciprocity
- Heterogeneity in the degree-related distributions
- Triadic and local structures
- Node attribute effects
  - The effect of a node-level attribute on the indegree (*alter* effects) and outdegree (*ego* effects) distribution.
  - Ex.: **Gender**. A *male alter* effect tests if male nodes are more likely to receive ties, while the *male ego* effect tests if male nodes are more likely to send ties.

How to the specify the effects for an SAOM?

- Outdegree (density) and reciprocity
- Heterogeneity in the degree-related distributions
- Triadic and local structures
- Node attribute effects
- Dyadic effects
  - The effect of a dyad-level covariate on the formation/creation of an $i \rightarrow j$ tie.
  - Ex.: same gender, same class, same neighborhood, socio-economic status similarity.

How to the specify the effects for an SAOM?

Look at the RSiena manual for a complete list of possible effects!

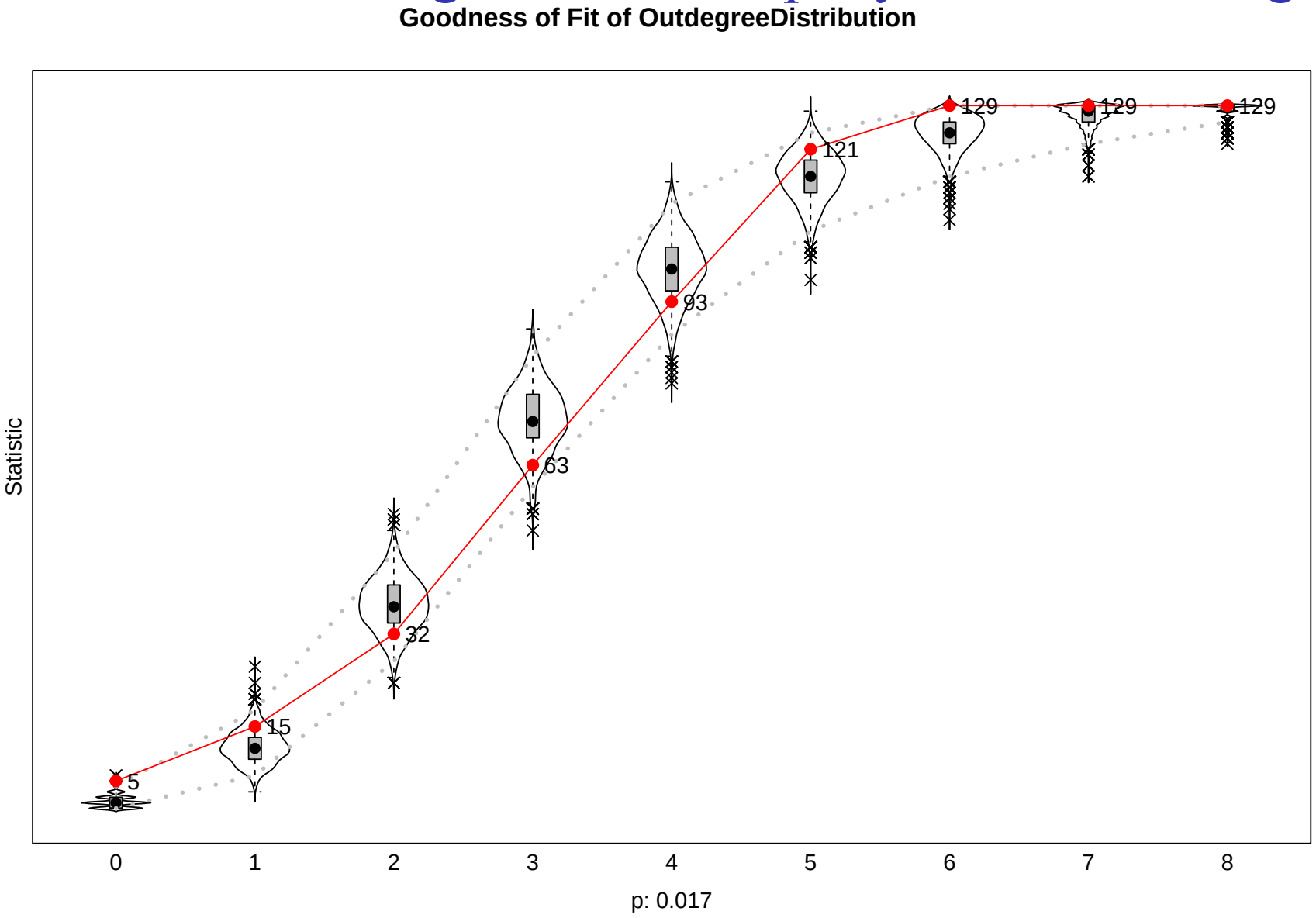


Core considerations for estimating a model

Model goodness-of-fit (GOF) [Lospinoso and Snijders, 2019]:

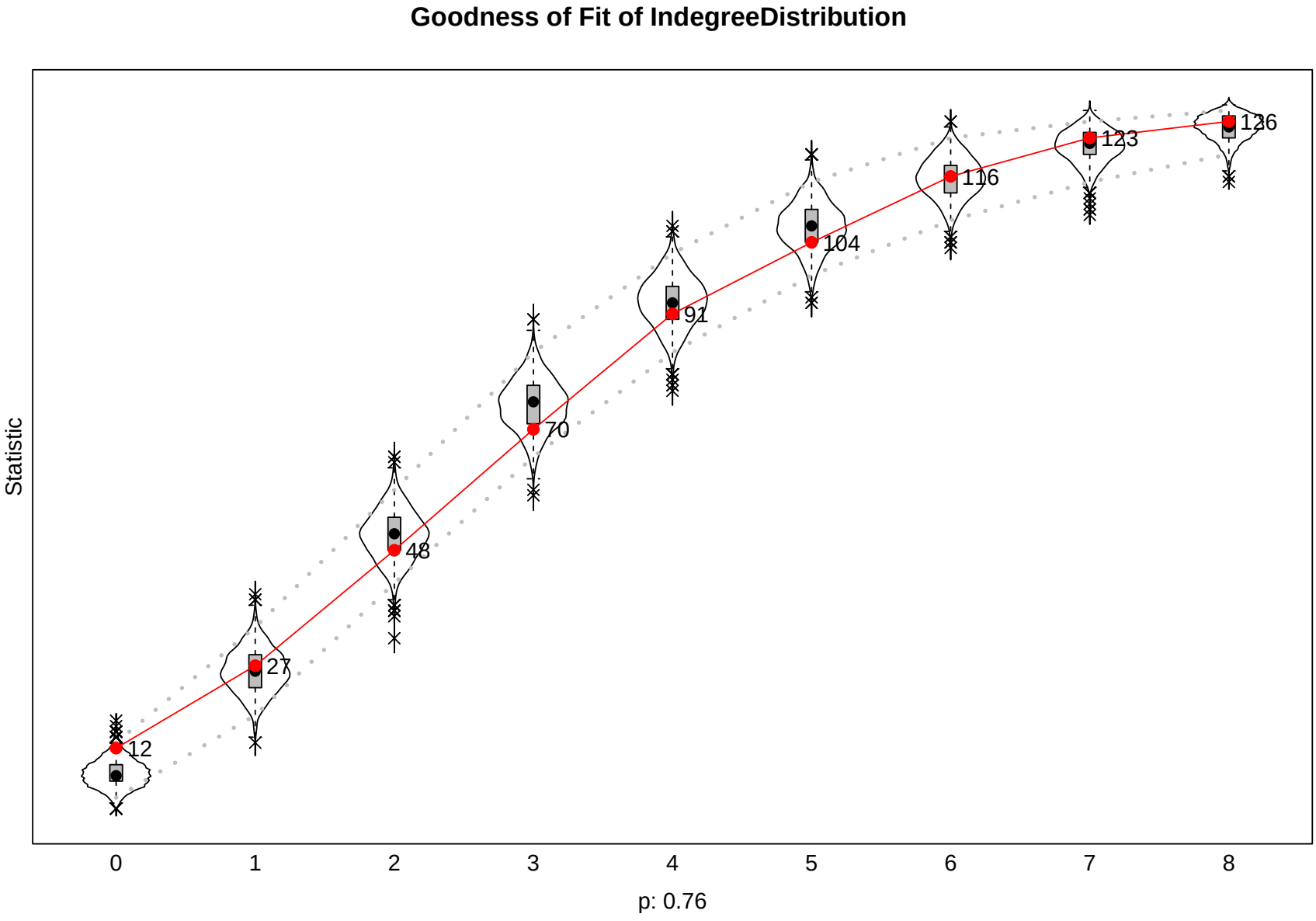
- Model goodness-of-fit: how well does the fitted model capture the underlying data generating process?
- Following Lospinoso and Snijders [2019], the SAOM GOF is assessed by generating a set of simulated networks for wave 2 based upon the fitted model and checking the Mahalanobis distance for each computed statistic.
- Ex. the indegree distribution: compare the count of the number of actors with indegree ($x_{+j} = 3$) in the simulated networks to the number of actors with indegree equal to 3 in the observed network.
- Mahalanobis distance p-values are computed for each distribution with smaller p-values indicating worse GOF.

GOF Results of Glasgow Friendship Dynamics: Outdegree



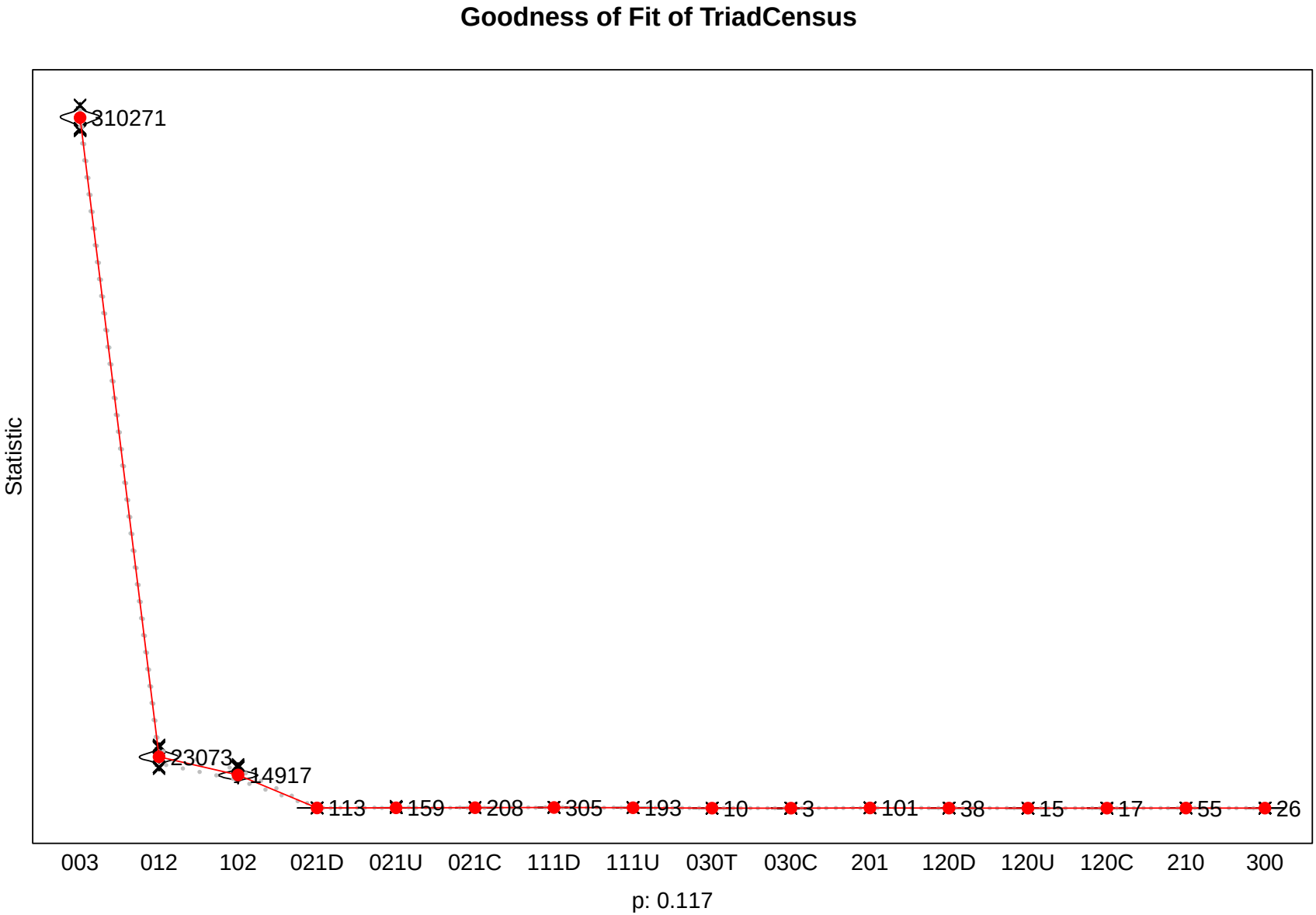
Mahalanobis distance p-value: 0.017 (We are doing okay [underfitting isolates]!)

GOF Results of Glasgow Friendship Dynamics: Indegree



Mahalanobis distance p-value: 0.76 (We are doing pretty good!)

GOF Results of Glasgow Friendship Dynamics: Triad Census



Mahalanobis distance p-value: 0.117 (We are doing good!)

Core considerations for estimating a model

- Model goodness-of-fit
- Time heterogeneity (for more than two waves)
  - An earlier assumption in continuous-time Markov models is *time homogenity* in the process, s.t. the effects of the network structure is constant across time.
    - $\forall_{t \in R} \beta_k(t) = \beta_k.$
  - Lospinoso et al. [2011] provide a method for assessing time heterogeneity when there are more than two waves of data collection!
  - How to fix time heterogeneity? (a) split the analyses by period or (b) include an interaction between the period and the network effect [Lospinoso et al., 2011].

Core considerations for estimating a model

- Model goodness-of-fit
- Time heterogeneity (for more than two waves)
- Model misspecification [Block, 2014, Leal et al., 2026]
  - Do the effects differ for tie creation and tie maintenance?
  - Ex. Is gender homophily 'more important' for maintaining the $i \rightarrow j$ tie than creating the $i \rightarrow j$ tie?
  - Block [2014, 97-119] and Leal et al. [2026] show that the standard assumption that the effects are the same for both processes can lead to null and/or disparate findings when the effect differs for tie creation and maintenance!
    - Without loss of generality, if $\beta_{\text{create}} > 0$ and $\beta_{\text{maintain}} < 0$, then the standard tie evaluation effect will lead to findings of a null effect.

Core considerations for estimating a model

- Model goodness-of-fit
- Time heterogeneity (for more than two waves)
- Model misspecification
- Missing data
  - Like in standard regression analysis, missing data are problematic for the SAOM!
  - Two types of missing data: (a) missing node attribute data and (b) missing tie data.
  - Look at the RSiena manual for different procedures for handling missing data! As Duxbury [2023] points out, list-wise deletion should be the last option for handling missing network data.

Core considerations for estimating a model

- Model goodness-of-fit
- Time heterogeneity (for more than two waves)
- Model misspecification
- Missing data
- Composition changes
  - Actors may enter and leave during process.
  - The SAOM provides the *structural zeros* and the *method of joiners and leavers* options to control for this.

Core considerations for estimating a model

- Model goodness-of-fit
- Time heterogeneity (for more than two waves)
- Model misspecification
- Missing data
- Composition changes
- Stochastic simulations
  - The SAOM finds the set of parameter estimates with an agent-based simulation approach, with means that the estimated values will vary over estimation runs.
  - One can set the random number seed to get the same results across simulations, but it is good practice to check the estimated values across different seeds (e.g., 37, 3718, 9999).

Core considerations for estimating a model

- Model goodness-of-fit
- Time heterogeneity (for more than two waves)
- Model misspecification
- Missing data
- Composition changes
- Statistical power
  - The statistical power, $\pi(g, W)$, for detecting significant effects is dependent upon g and the number of waves. Let $\pi(g, W) = \text{P(Reject } H_0 | \beta_1 \neq 0)$
  - In a simulation study, Stadtfeld et al. [2020] found that, for example, the statistical power for dyadic homophily effects:
    - when $g = 20$ and $W = 2$, $\pi(g, W) = 0.345$.
    - when $g = 120$ and $W = 2$, $\pi(g, W) = 0.995$.
    - when $g = 60$ and $W = 3$, $\pi(g, W) = 0.995$.

Post-Estimation Simulations

Simulations in longitudinal network analysis are powerful research tools:

- → Micro-effects on Macro Network Structures [Duxbury, 2024]
  - ☞ What is the effect of reciprocity on network density?
  - ☞ How does gender homophily impact network clustering?
- → Network-based Intervention Strategies [Duxbury and Haynie, 2020]
  - ☞ What would happen to the rate of smoking in a high school network if the most popular students promoted smoking cessation?

Post-Estimation Simulations: How does gender homophily impact network clustering?

Micro-effects on Macro Network Structures [Duxbury, 2024]

- → Network clustering defined as the graph transitivity value, which is the ratio of the number of closed triangles to the number of connected triples (ranges from 0 to 1).

- → Denote $g(A)$ be the graph transitivity value for a sociomatrix A. Given a fitted SAOM, we can generate simulated sociomatrices, $\hat{A}$, based upon the estimated parameter values.

- → Let $\hat{A}(x; \hat{\beta}_{SG})$ be a simulated sociomatrix when the effect of gender homophily is set to it's estimated value and $\hat{A}(x; 0)$ be a simulated sociomatrix when the effect of gender homophily is set to 0.

Post-Estimation Simulations: How does gender homophily impact network clustering?

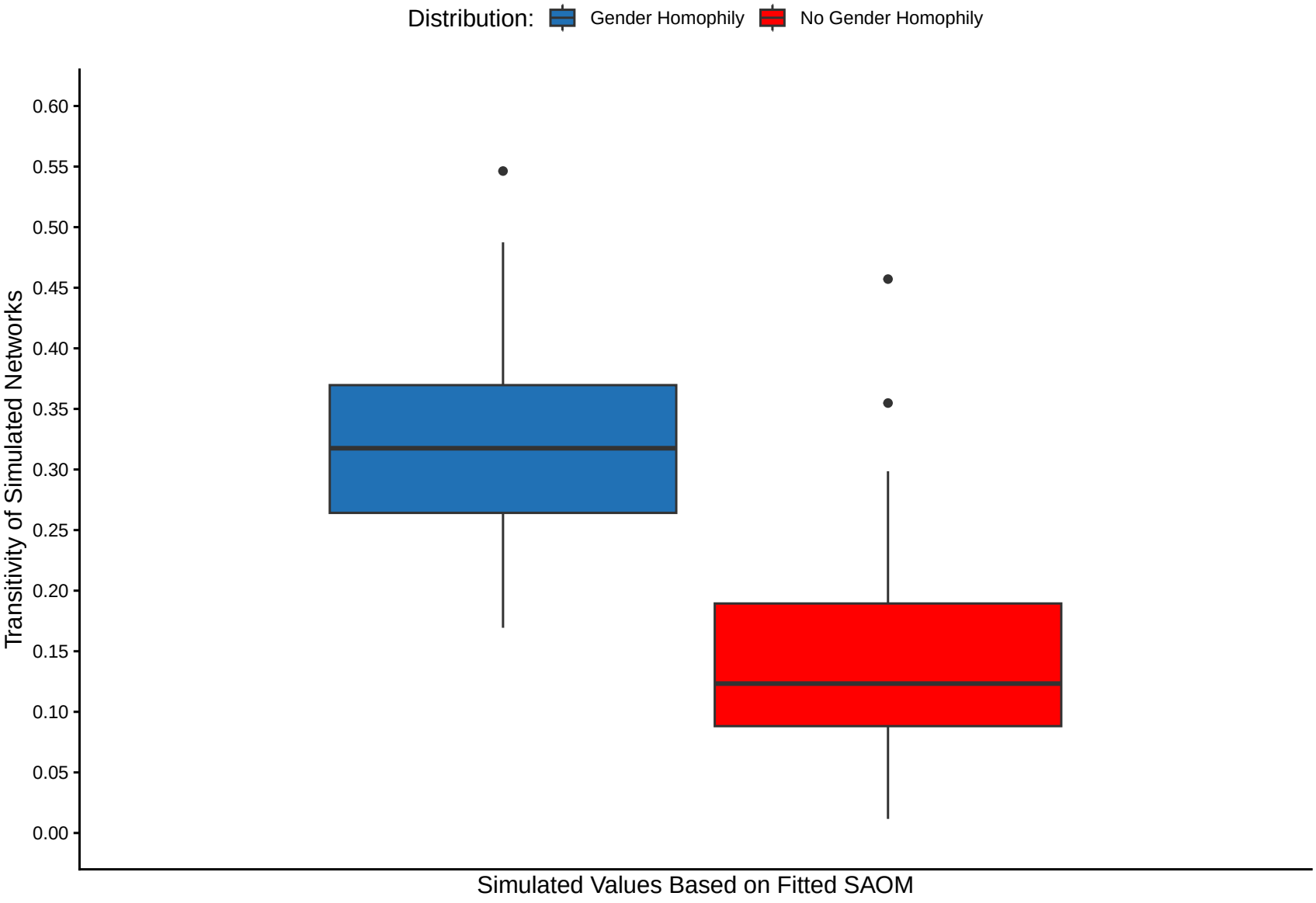
Micro-effects on Macro Network Structures [Duxbury, 2024]

→ The micro-effect on macro network structures for gender homophily on network clustering is thus:

$$MEMS_{SG} = g(\hat{A}(x; \hat{\beta}_{SG})) - g(\hat{A}(x; 0))$$

→ Replicate the stochastic procedure 100 times to generate a $MEMS$ distribution.

How does gender homophily impact network clustering?



$$\hat{MEMS} = 0.1763 \text{ (SD : } 0.093;\ p\text{-value} = 0.058;\ \text{MC-}p = 0.03)$$

Post-Estimation Simulations: How does gender homophily impact network clustering?

We can also use the **netmediate** R package to estimate the MEMS for our empirical application!

```
MEMS(saom.fit,
     micro_process = "same gender",
     macro_function = sna::gtrans,
     nsim = 100,
     SAOM_data=siena.data)
```

	Estimate	Std. Dev.	lower 95% CI	Upper 95% CI	MC p-val
MEMS	0.163	0.084	0.012	0.317	0.02
Prop. Change in M	0.515	NA	NA	NA	NA

Concluding remarks

The **stochastic actor-oriented model (SAOM)** is an inferential continuous-time markov model that allows us to understand how a network changes across waves of data collection [Snijders, 1996, Snijders et al., 2010].

We have looked over model interpretation, checking goodness-of-fit, and conducting post-estimation simulations!

Next week, we will discuss **exponential random graph models** (ERGMs)!



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